

一、

1. 样本函数: $X(t) = t + 2$; $X(t) = -t$; $X(t) = t - 2$

自协方差函数: $C_X(t, s) = \text{Cov}[X(t), X(s)]$

$$= \text{Cov}[ABt + A + B, ABs + A + B]$$

$$= \text{Cov}[ABt, ABs] + \text{Cov}[ABt, A] + \text{Cov}[ABt, B]$$

$$+ \text{Cov}[A, ABs] + \text{Cov}[A, A] + \text{Cov}[A, B] + \text{Cov}[B, ABs]$$

$$+ \text{Cov}[B, A] + \text{Cov}[B, B]$$

$$= ts [E[A^2]E[B^2] - (E[A]E[B])^2] + 0 + 1 + 1$$

$$= ts + 2$$

$$P(X(2) \geq 0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$2. S_X(\omega) = F\{R_X(\tau)\} = 2\pi \delta(\omega) + 1$$

③ 正态

$$e^{\frac{1}{2}t} \star E[e^Y] = e^{(\mu + \frac{\sigma^2}{2})}, \text{其中 } Y \sim N(\mu, \sigma^2)$$

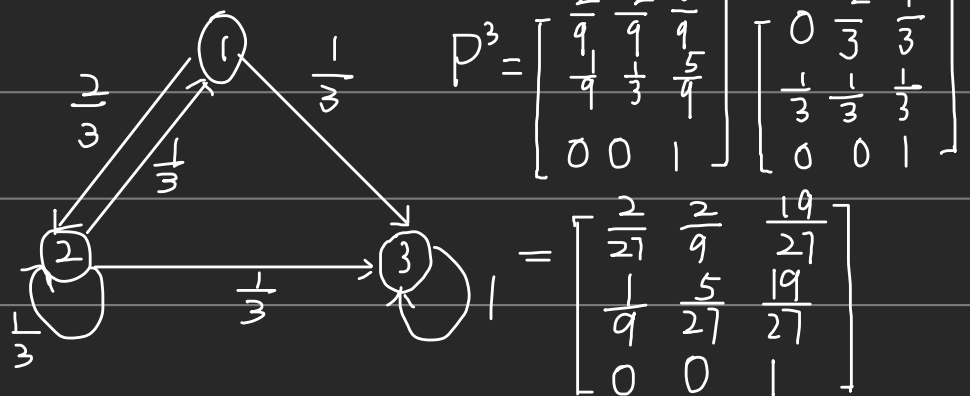
$$\begin{aligned} \textcircled{4.11} P(X(3)=4 | X(1) \geq 2) &= \frac{P(X_1=2)}{P(X_1 \geq 2)} P(X(3)-X(1)=2) \\ &+ \frac{P(X_1=3)}{P(X_1 \geq 2)} P(X(3)-X(1)=1) + \frac{P(X_1=4)}{P(X_1 \geq 2)} P(X(3)-X(1)=0) \\ &= \frac{\frac{e^{-1}}{2} e^{-2 \frac{2^2}{2}} + \frac{e^{-1}}{3!} e^{-2 \cdot 2} + \frac{e^{-1}}{4!} e^{-2}}{1 - 2e^{-1}} \\ &= \frac{33}{24e^3 - 48e^2} \end{aligned}$$

$$\begin{aligned}
 (2) P(X(4) + Y(2) = 2) &= P(X(4) = 2)P(Y(2) = 0) + P(X(4) = 1, Y(2) = 1) \\
 &\quad + P(X(4) = 0)P(Y(2) = 2) \\
 &= e^{-4} \frac{4^2}{2!} e^{-4} \frac{4^0}{0!} + e^{-4} \frac{4^1}{1!} e^{-4} \frac{4^1}{1!} + e^{-4} \frac{4^0}{0!} e^{-4} \frac{4^2}{2!} \\
 &= 32e^{-8}
 \end{aligned}$$

5.

$$(1) P^2 = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} \frac{2}{9} & \frac{2}{9} & \frac{5}{9} \\ \frac{1}{9} & \frac{1}{3} & \frac{5}{9} \\ 0 & 0 & 1 \end{bmatrix}$$

$$P(X_2 = 2) = \frac{1}{2} \times \frac{2}{9} + \frac{1}{2} \times \frac{1}{3} = \frac{5}{18}$$



$$\begin{aligned}
 (2) P(X_0 = 2, X_1 = 2, X_3 = 3) &= P(X_0 = 2) P_{22}^{(1)} P_{23}^{(2)} \\
 &= \frac{1}{2} \times \frac{1}{3} \times \frac{5}{9} = \frac{5}{54}
 \end{aligned}$$

$$(3) P(X_3 = 3) = \frac{1}{2} P_{13}^{(3)} + \frac{1}{2} P_{23}^{(3)} = \frac{1}{2} \times \frac{19}{27} + \frac{1}{2} \times \frac{19}{27} = \frac{19}{27}$$

$$(4) P(X_1 = 1 | X_2 = 2) = \frac{P(X_1 = 1, X_2 = 2)}{P(X_2 = 2)} = \frac{\frac{1}{6} \times \frac{2}{3}}{\frac{5}{18}} = \frac{2}{5}$$

(5)

$$\begin{cases} m_1 = 1 + \frac{2}{3}m_2 + \frac{1}{3}m_3 \\ m_2 = \frac{1}{3}m_1 + \frac{1}{3}m_2 + \frac{1}{3}m_3 + 1 \\ m_3 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} m_1 = 3 \\ m_2 = 3 \end{cases}$$

$$E(\tau) = 3 \times \frac{1}{2} + 3 \times \frac{1}{2} = 3$$

二、(1)

$$\begin{aligned} 2B(1) - B(2) &= 2B(1) - (B(2) - B(1) + B(1)) \\ &= B(1) - (B(2) - B(1)) \end{aligned}$$

$$\therefore 2B(1) - B(2) \sim N(0, 2)$$

$$(2) \quad \text{令 } \tilde{B}(t) = \begin{cases} tB(1/t), & t > 0 \\ 0, & t = 0 \end{cases} \text{, 为标准布朗运动}$$

$X_s(t) = \tilde{B}(t+1) - B(1) = (t+1)B(\frac{1}{t+1}) - 1$, 也为标准布朗运动

$$(3) \quad P\{B(1) > 1 \mid B(2) = 2\}$$

$$\text{令 } \tilde{B}(t) = \begin{cases} tB(1/t), & t > 0 \\ 0, & t = 0 \end{cases}$$

$$\begin{aligned}\therefore P\{B(1) > 1 \mid B(2) = 2\} &= P\{\tilde{B}(1) > 1 \mid \tilde{B}(\frac{1}{2}) = 1\} \\ &= P(\tilde{B}(1) - \tilde{B}(\frac{1}{2}) > 0) = \frac{1}{2}\end{aligned}$$

$$\begin{aligned}(4) P\{\min_{1 \leq t \leq 2} B(t) \leq -2 \mid B(1) = -1\} \\ = P\{\min_{1 \leq t \leq 2} B(t) - B(1) \leq -1\}\end{aligned}$$

$$= P\{\min_{0 \leq s \leq 1} B(s) \leq -1\} = 2\Phi(-1) = 0.32.$$

(5) X_t 为布朗桥运动, $D_X(t) = t(1-t)$

三、

$$\begin{aligned}(1). M_X(t) &= E[\cos(t+\Theta) + e^{-t} B(e^{2t})] \\ &= \frac{1}{2\pi} \int_0^{2\pi} \cos(t+\Theta) d\Theta + e^{-t} E[B(e^{2t})] \\ &= 0 + 0 = 0\end{aligned}$$

$$\begin{aligned}C_X(t,s) &= \text{Cov}[X(t), X(s)] = \text{Cov}[\cos(t+\Theta) + e^{-t} B(e^{2t}), \\ &\quad \cos(s+\Theta) + e^{-s} B(e^{2s})] \\ &= \text{Cov}[\cos(t+\Theta), \cos(s+\Theta)] + \text{Cov}[e^{-t} B(e^{2t}), e^{-s} B(e^{2s})]\end{aligned}$$

$$\begin{aligned}
 &= E[\cos(t+\theta)\cos(s+\theta)] - E[\cos(t+\theta)]E[\cos(s+\theta)] \\
 &\quad + e^{-(t+s)} \min\{e^{2t}, e^{2s}\} \\
 &= \frac{1}{2}\cos(t-s) + e^{-(t+s)} \min\{e^{2t}, e^{2s}\}
 \end{aligned}$$

$$\begin{aligned}
 \therefore R_X(t,s) &= C_X(t,s) + \mu_X(t)\mu_X(s) \\
 &= \frac{1}{2}\cos(t-s) + e^{-|t-s|}
 \end{aligned}$$

$\therefore \mu_X(t)$ 为常值, $R_X(t,s)$ 为时间差 $t-s$ 的函数

$\therefore$ 为宽平稳过程

(2) 由(1), $R_X(\tau) = \frac{1}{2}\cos\tau + e^{-|\tau|}$

$\lim_{\tau \rightarrow +\infty} (\frac{1}{2}\cos\tau + e^{-|\tau|})$ 的极限不存在

故不具有各态历经性

无平均

(3) $S_X(\omega) = \frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{2}{\omega^2 + 1}$

四、(1) 强度为 5 的泊松分布

(2) $P(N_1=0, N_2=4 \text{ 或 } N_1=1, N_2=3 | N_1+N_2=4)$

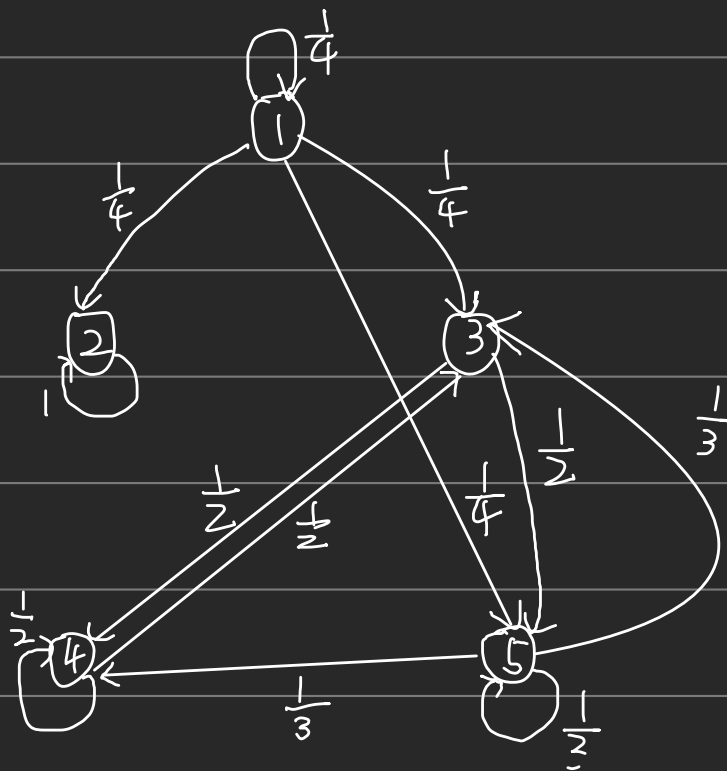
$$= \frac{e^{-2} e^{-3} \frac{3^4}{4!} + e^{-2} \frac{2^1}{1!} e^{-3} \frac{3^3}{3!}}{e^{-5} \frac{5^4}{4!}}$$

$$= \frac{\frac{81}{24} + \frac{216}{24}}{\frac{625}{24}} = \frac{297}{625}$$

$$(3) P(N_1=0, N_2=1 | N_1+N_2=1) \\ = \frac{e^{-2} e^{-3} \frac{3^1}{1!}}{e^{-5} \frac{5}{1!}} = \frac{3}{5}$$

$$(4) P(N_1(1)=0, N_2(2) \geq 2) + P(N_1(1)=1, N_2(2) \geq 2) \\ = e^{-3} [1 - e^{-3} - e^{-3} \frac{3^1}{1!}] + e^{-3} \frac{3^1}{1!} [1 - e^{-3}] \\ = e^{-3} [1 - e^{-3} - 3e^{-3} + 3 - 3e^{-3}] = e^{-3} (4 - 7e^{-3})$$

五.



(1) $\{1\}, \{2\}, \{3, 4, 5\}$, 后两个是闭的

(2) $d_1 = d_2 = d_3 = d_4 = d_5 = 1$

2, 3, 4, 5 正常返, 1 暂留

$$(3) \quad \mu_2 = 1,$$

$$(\pi_3, \pi_4, \pi_5) \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} = (\pi_3, \pi_4, \pi_5)$$

$$\Rightarrow \begin{cases} \pi_3 = \frac{4}{13} \\ \pi_4 = \frac{6}{13} \\ \pi_5 = \frac{3}{13} \end{cases} \quad \therefore \begin{cases} \mu_3 = \frac{13}{4} \\ \mu_4 = \frac{13}{6} \\ \mu_5 = \frac{13}{3} \end{cases}$$

(4) 设 $h_i = P(\text{从 } i \text{ 出发, 被 2 吸收})$

$$\therefore \begin{cases} h_1 = \frac{1}{4}h_1 + \frac{1}{4}h_2 + \frac{1}{4}h_3 + \frac{1}{4}h_5 \\ h_2 = 1 \\ h_3 = h_4 = h_5 = 0 \end{cases}$$

$$\Rightarrow h_1 = \frac{1}{3}$$

$$\therefore \lim_{n \rightarrow \infty} P(X_n = 2) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

设 $p_i = P(\text{从 } i \text{ 出发, 被 } \{3, 4, 5\} \text{ 吸收})$

$$p_1 = \frac{1}{4}p_1 + 0 + \frac{1}{2} \Rightarrow p_1 = \frac{2}{3}$$

$$\therefore \lim_{n \rightarrow \infty} P(X_n = 3) = \left(\frac{1}{2} + \frac{1}{2} \times \frac{2}{3}\right) \times \frac{4}{13} = \frac{10}{39}$$

